

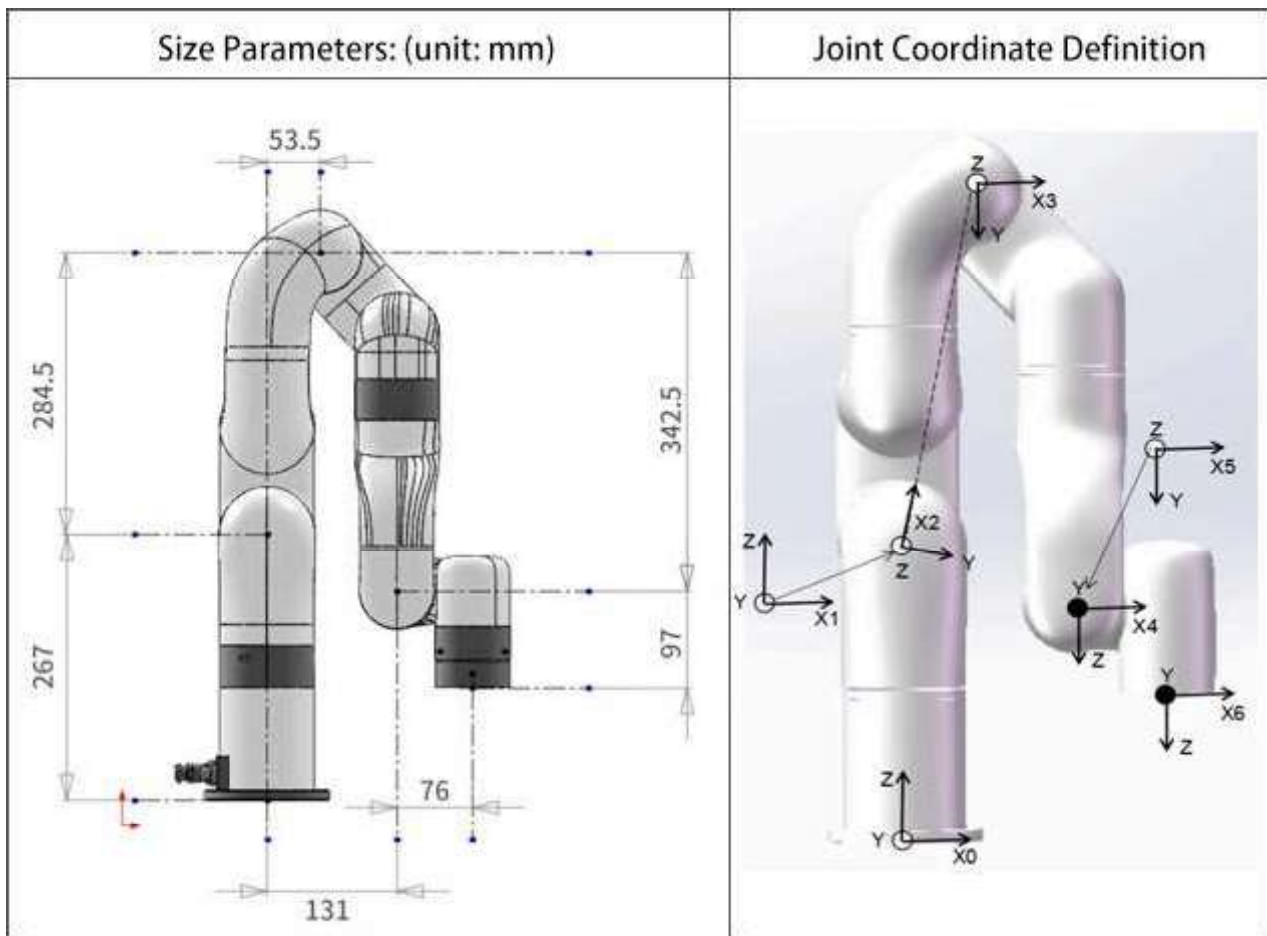
Dynamics	Mass (Kg)	Center of Mass (mm)
Link1	2.46	[0.13, 30.1, -12.0]
Link2	2.21	[38.2, -226.6, 34.7]
Link3	2.16	[69.0, 231.8, 1.0]
Link4	1.354	[65.2, 31.8, 3.11]
Link5	0.17	[0, -6.77, -10.98]

UFACTORY xArm 5-Model 3

Dynamics	Mass (Kg)	Center of Mass (mm)
Link1	2.382	[0.13, 29.4, -12.4]
Link2	2.164	[37.88, -225.4, 34.47]
Link3	2.121	[68.83, 229.85, 1.02]
Link4	1.317	[64.9, 31.2, 3.2]
Link5	0.17	[0, -6.77, -10.98]

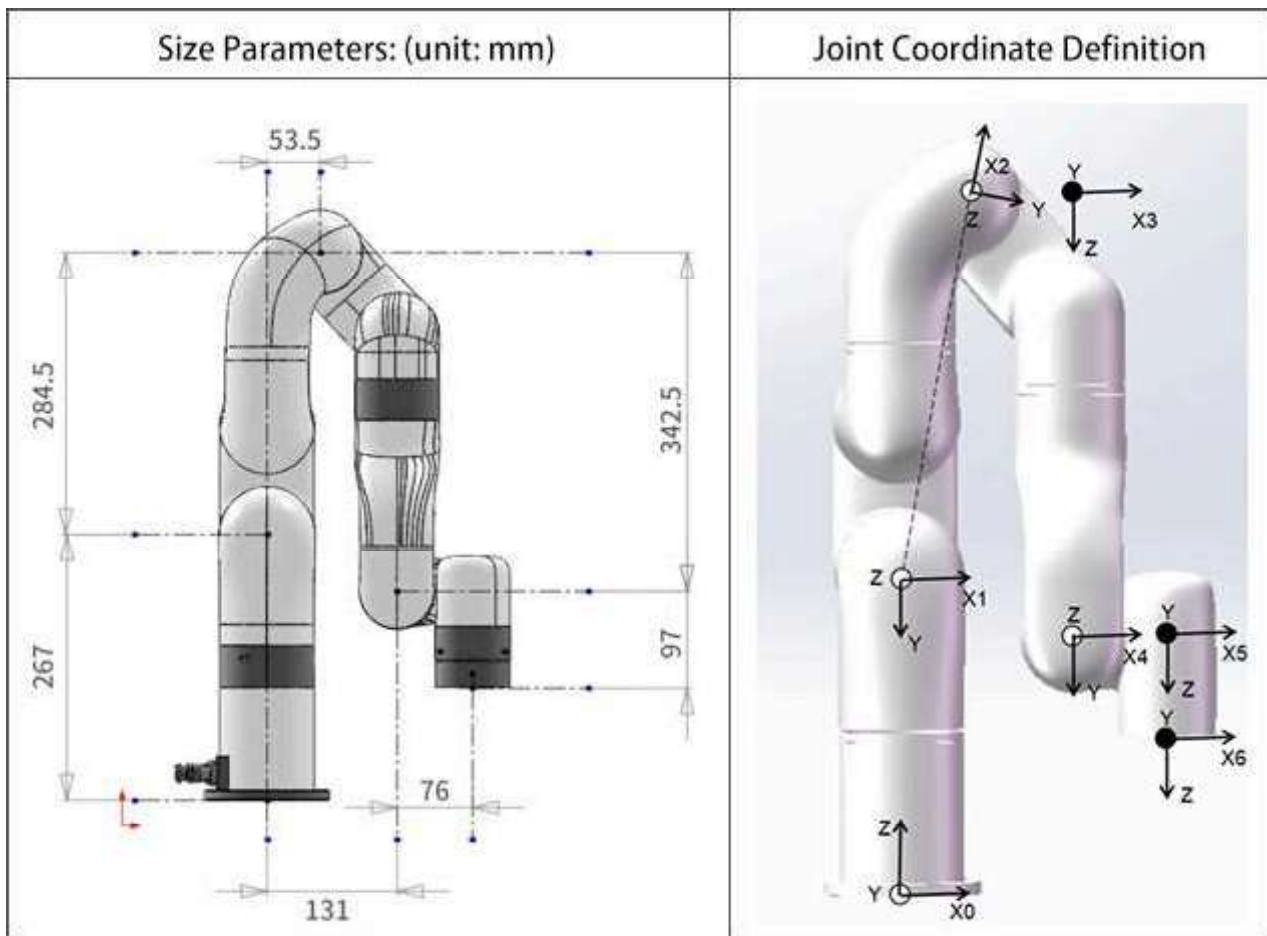
2. UFACTORY xArm 6

1) UFACTORY xArm 6 Modified D-H Parameters



Kinematics	theta (rad)	d (mm)	alpha (rad)	a (mm)	offset (rad)
Joint1	0	267	0	0	0
Joint2	0	0	-pi/2	0	T2_offset
Joint3	0	0	0	a2	T3_offset
Joint4	0	342.5	-pi/2	77.5	0
Joint5	0	0	pi/2	0	0
Joint6	0	97	-pi/2	76	0

2) UFACTORY xArm 6 Standard D-H Parameters



Kinematics	theta (rad)	d (mm)	alpha (rad)	a (mm)	offset (rad)
Joint1	0	267	-pi/2	0	0
Joint2	0	0	0	a2	T2_offset
Joint3	0	0	-pi/2	77.5	T3_offset
Joint4	0	342.5	pi/2	0	0
Joint5	0	0	-pi/2	76	0
Joint6	0	97	0	0	0

Note:

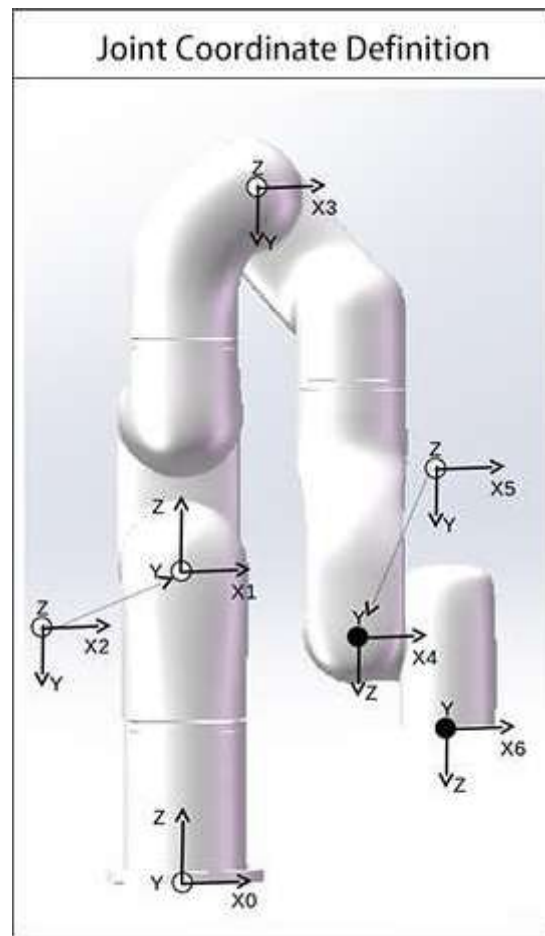
'**Tx_offset**' is the offset joint angle from the mathematical zero position to the mechanical zero position shown in the picture.

$$a2 = \sqrt{284.5^2 + 53.5^2} = 289.48866;$$

$$T2_offset = -\text{atan}(284.5/53.5) = -1.3849179 \text{ } (-79.34995^\circ);$$

$$T3_offset = -T2_offset = 1.3849179 \text{ } (79.34995^\circ);$$

3) UFACTORY xArm 6 Mass Parameters



UFACTORY xArm 6-Model 1

Dynamics	Mass (Kg)	Center of Mass (mm)
Link1	2.177	[0.15, 27.24, -13.57]
Link2	2.011	[36.7, -220.9, 33.56]
Link3	1.725	[69.77, 113.5, 11.6]
Link4	1.211	[-0.2, 20.0, -26.0]
Link5	1.206	[63.87, 29.3, 3.5]
Link6	0.17	[0, -6.77, -10.98]

UFACTORY xArm 6-Model 2

Dynamics	Mass (Kg)	Center of Mass (mm)
Link1	2.46	[0.13, 30.1, -12.0]
Link2	2.21	[38.2, -226.6, 34.7]
Link3	1.925	[70.6, 117.2, 10.4]
Link4	1.36	[0.18, 17.7, -23.0]
Link5	1.354	[65.2, 31.8, 3.11]
Link6	0.17	[0, -6.77, -10.98]

UFACTORY xArm 6-Model 3

Dynamics	Mass (Kg)	Center of Mass (mm)
Link1	2.382	[0.13, 29.4, -12.4]
Link2	2.267	[38.8, -227.8, 34.96]
Link3	1.875	[70.4, 116.3, 10.7]
Link4	1.319	[-0.2, 18.0, -22.9]
Link5	1.34	[65.1, 30.96, 3.15]
Link6	0.17	[0, -6.77, -10.98]

3. UFACTORY xArm 7

1) UFACTORY xArm7 Modified D-H Parameters